

Dsp EEE | ECE F434Tut-1Introduction

1.) System properties

a) i) $y(n) = Ax(n) + B$
 ii) $y(n) = x(n^2)$

} Check whether the system described by these equations is linear or non-linear

Given $x(n) \rightarrow \text{I/p}$, $y(n) \rightarrow \text{O/p}$.

b) Given $x(n) = \begin{cases} 3(-1)^n; & n \geq 0 \\ 0; & n < 0 \end{cases}$. Is this signal energy or power signal.

2.) Find the cross correlation sequence of given sequences

$$x(n) = \{\dots, 0, 0, 2, -1, 3, 7, \underset{\uparrow}{1}, 2, -3, 0, 0, \dots\}$$

$$y(n) = \{\dots, 0, 0, 1, -1, 2, -2, \underset{\uparrow}{4}, 1, -2, 5, 0, 0, \dots\}$$

Discuss MATLAB implementation.

3.) Sequence of Fibonacci numbers is a causal sequence defined by

$$f[n] = f[n-1] + f[n-2]; \quad n \geq 2; \text{ with } f[0] = 0 \text{ and } f[1] = 1.$$

a) develop an exact formula to calculate $f[n]$ directly for any 'n'.

b) Show that $f[n]$ is IR of a causal LTI system described by $y[n] = y[n-1] + y[n-2] + x[n-1]$

4.) Let $x(n)$ & $y(n)$ be two independent zero mean WSS random signals with autocorrelations $R_{xx}(n)$ & $R_{yy}(n)$. Consider a random signal $v(n)$ obtained as linear combination of $x(n)$ & $y(n)$: $v(n) = ax(n) + by(n)$ where 'a' & 'b' constants. Express auto & cross correlations $R_{vv}(n)$, $R_{vx}(n)$, $R_{vy}(n)$ in terms of $R_{xx}(n)$ & $R_{yy}(n)$.

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